



HK IMO Training 2023-2024 Problem Set 16 Suggested Solutions



Problem 61 (Thailand online MO 2023 Problem 3). Let m and n be positive integers satisfying $(m, n!) = 1$. Prove that $n! \mid m^{n!} - 1$.

Solution. The result is obvious if $m = 1$ or $n = 1$. Suppose $m, n \geq 2$. For any odd prime divisor p of $n!$, we must have $p \leq n$ and $p \nmid m$. Thus, $p - 1 \leq n$ and hence $p - 1 \mid n!$. By the Fermat little theorem, we have $p \mid m^{p-1} - 1$. Therefore, by the lifting the exponent lemma,

$$v_p(m^{n!} - 1) = v_p((m^{p-1})^{\frac{n!}{p-1}} - 1) = v_p(m^{p-1} - 1) + v_p\left(\frac{n!}{p-1}\right) > v_p\left(\frac{n!}{p-1}\right) = v_p(n!)$$

as $(p-1, p) = 1$. Similarly, as m is odd,

$$v_2(m^{n!} - 1) = v_2((m^2)^{\frac{n!}{2}} - 1) = v_2(m^2 - 1) + v_2\left(\frac{n!}{2}\right) \geq 3 + v_2(n!) - 1 > v_2(n!).$$

Therefore, we must have $n! \mid m^{n!} - 1$. □

Problem 62 (Thailand MO 2021 Problem 3). Let x, y, z be positive real numbers satisfying $xy + yz + zx = xyz$. Find the minimum value of

$$x^x yz + xy^y z + xyz^z.$$

Solution. The minimum value is 729.

The given condition is the same as

$$\frac{1}{x} + \frac{1}{y} + \frac{1}{z} = 1.$$

By the weighted AM-GM inequality, we have

$$\frac{1}{x}(x^x) + \frac{1}{y}(y^y) + \frac{1}{z}(z^z) \geq \left(\frac{1}{x} + \frac{1}{y} + \frac{1}{z}\right) ((x^x)^{\frac{1}{x}} (y^y)^{\frac{1}{y}} (z^z)^{\frac{1}{z}})^{\frac{1}{\frac{1}{x} + \frac{1}{y} + \frac{1}{z}}} = xyz.$$

This implies $x^{x-1} + y^{y-1} + z^{z-1} \geq xyz$, and hence

$$x^x yz + xy^y z + xyz^z = xyz(x^{x-1} + y^{y-1} + z^{z-1}) \geq (xyz)^2.$$

Now, by the AM-GM inequality,

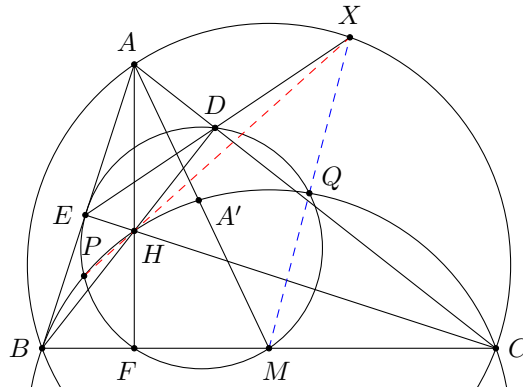
$$xyz = xy + yz + zx \geq 3\sqrt[3]{(xyz)^2}.$$

This implies $xyz \geq 27$. Therefore,

$$x^x yz + xy^y z + xyz^z \geq (xyz)^2 \geq 729.$$

Equality holds when $x = y = z = 3$. □

Problem 63 (Canada MO 2016 Problem 5). In $\triangle ABC$, D and E are the feet of altitudes from B and C respectively, H is the orthocentre, and M is the midpoint of BC . The circumcircles of $\triangle DEM$ and $\triangle BCH$ intersect at P and Q , with P being on the same side of the line AH as B . Prove that the lines DE , PH , MQ are concurrent at a point on the circumcircle of $\triangle ABC$.



Solution. Let X be the intersection point of the line DE and (ABC) such that D lies between E and X .

Firstly, consider the inversion at centre H that interchanges B and D . As B, C, D, E are concyclic, this inversion interchanges C and E . Similarly, A is mapped to the foot of altitude F of $\triangle ABC$. Therefore, (ABC) is mapped to (DEF) , while DE is mapped to (BCH) . Note that (DEF) is the nine-point circle, so it coincides with (DEM) . It follows that X is mapped to an intersection point of (DEM) and (BCH) . From the configuration, X is mapped to P . This shows X, H, P are collinear.

Secondly, consider the inversion at centre M with radius $MB = MC = MD = ME$. Let A' be the image of A . Since

$$\angle BA'C = \angle BA'M + \angle MA'C = \angle MBA + \angle ACM = \angle CAB = \angle BHC,$$

the point A' lies on (BCH) . Therefore, (ABC) is mapped to $(A'BC) = (BCH)$, while DE is mapped to (DEM) . Again, X is mapped to an intersection point of (BCH) and (DEM) . From the configuration, X is mapped to Q . This shows X, Q, M are collinear.

Therefore, we have proved that DE, PH, MQ are concurrent at X . □

Problem 64 (All-Russian Olympiad 2021 Grade 11 Problem 8). There are 10000 coins in 100 colours, with exactly 100 coins of each colour. The coins are distributed to 100 people randomly such that each person receives exactly 100 coins. In each move, two people may exchange one of their coins. Suppose it is possible to exchange the coins finitely many times such that each person has 100 coins of different colours in the end. Prove that this can be done in a way that no coin is exchanged more than once.

Solution. We first prove the following result.

Claim. Suppose there are $100k$ coins in 100 colours, with exactly k coins of each colour. If we distribute these coins to 100 people randomly such that each person receives exactly k coins, then each person can select one of his coins such that the 100 coins selected by all people are of distinct colours.

Proof. Consider a bipartite graph $G = (V_1, V_2, E)$ where V_1 consists of the 100 people and V_2 consists of the 100 colours. An edge is drawn between a person v_i and a colour c_j if v_i has a coin of colour c_j . For any subset W of V_1 with $|W| = m$, the people in W have km coins in total. These coins are of at least $\frac{km}{k} = m$ different colours. Therefore, the vertices in W are adjacent to at least m vertices in V_2 . As this holds for all $W \subset V_1$, G has a perfect matching by Hall's theorem. WLOG assume $v_i c_i$ is an edge for each $i = 1, 2, \dots, 100$. Then person v_i can select a coin of colour c_i . □

Applying the claim with $k = 100, 99, \dots, 1$ step by step, we know that the people can rearrange their own coins in a certain order such that the k th coins of all 100 people are of different colours for each k . We use $a_{i,j}$ to denote the j th coin of a person v_i . This means $a_{1,j}, a_{2,j}, \dots, a_{100,j}$ are of different colours for each j . By exchanging the coins $a_{i,j}$ and $a_{j,i}$ for all pairs of distinct indices i, j , in the end $a_{j,1}, a_{j,2}, \dots, a_{j,100}$ are of different colours for each j . This means exactly that v_j has 100 coins of different colours as desired. □